

AI DESKTOP AUTOMATION AGENT

User & Instruction Manual

Local-first | Policy-guarded | Offline-capable | No paid APIs

Version 1.4.0 | October 2026

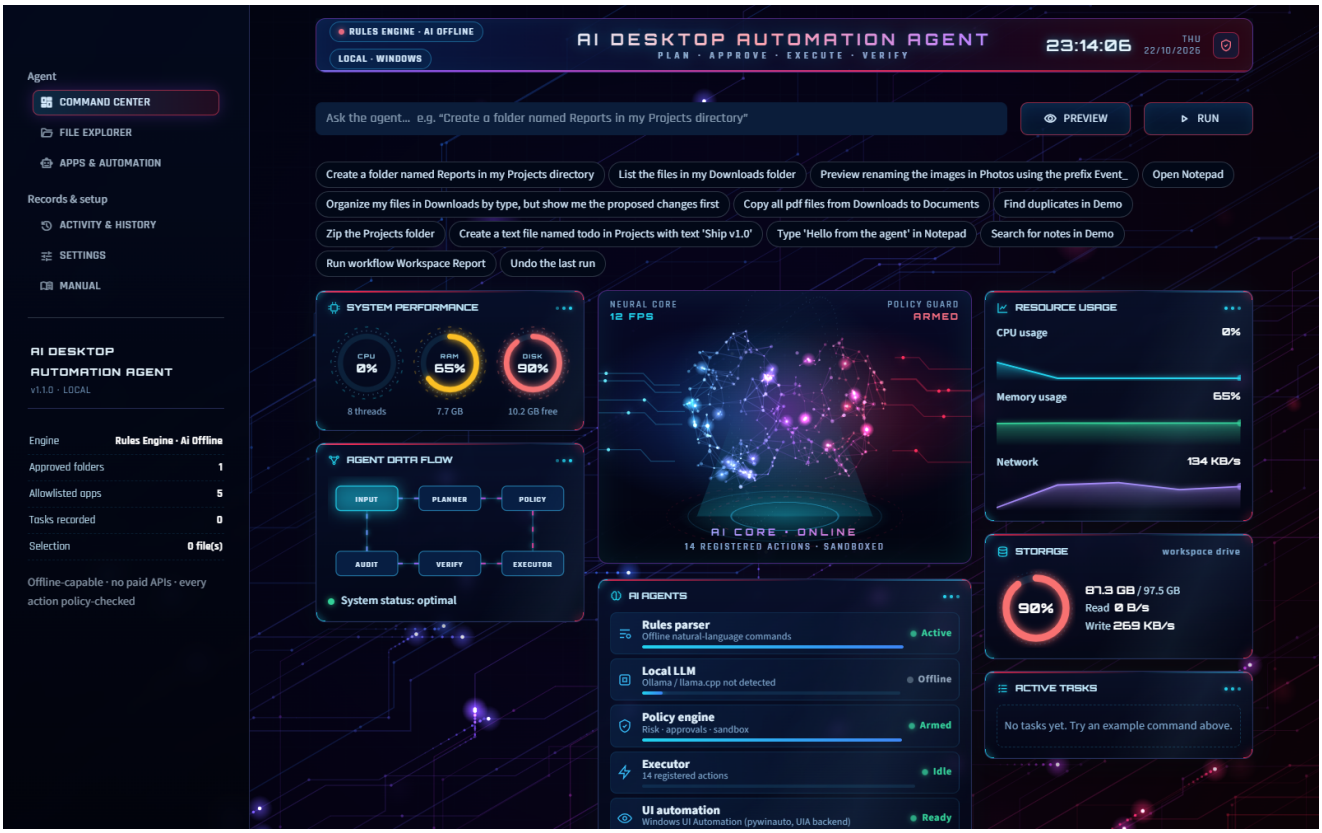
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1. What the agent does

The AI Desktop Automation Agent turns plain-English instructions into a visible, step-by-step plan, checks every step against a safety policy, asks for your approval when something sensitive is involved, then executes, verifies and records each action. It works fully offline with a built-in rules engine; a local AI model (Ollama or llama.cpp) is optional. No paid API, account or API key is ever needed.

- Files & folders: list, search, create, copy, move, rename, quarantine-delete, organize by type, find duplicates, zip, create text notes.
- Desktop: launch allowlisted apps, open approved files/folders, inspect accessible UI controls, type text into a window the agent opened (Windows UI Automation).
- Safety: approved-folder sandbox, risk levels, approval dialog, just-in-time confirmation, emergency STOP, dry runs, conflict detection, Undo, SQLite audit history.



Command Center: HUD dashboard, command bar and live agent status.

2. Requirements

- Windows 10 or 11 (desktop features). File features also work on macOS/Linux and in the hosted demo.
- Python 3.11 or newer from python.org (tick 'Add python.exe to PATH' during install).
- About 300 MB disk space for the virtual environment. No administrator rights needed.
- Optional: Ollama (ollama.com) or a llama.cpp server for freer phrasing. A small model such as llama3.2:3b needs roughly 2-3 GB disk and 4+ GB RAM.

3. Installation

- Unzip the project, e.g. to Desktop\AI Desktop Automation Agent\desktop-agent.
- Double-click setup.bat. It checks Python, creates a private .venv folder and installs the dependencies (Streamlit, Pydantic, psutil, ReportLab, pywinauto). Internet is needed once for this.
- If setup reports an error, read the message: the most common cause is Python missing from PATH.
- Nothing outside the project folder is changed, apart from the data folder described below.

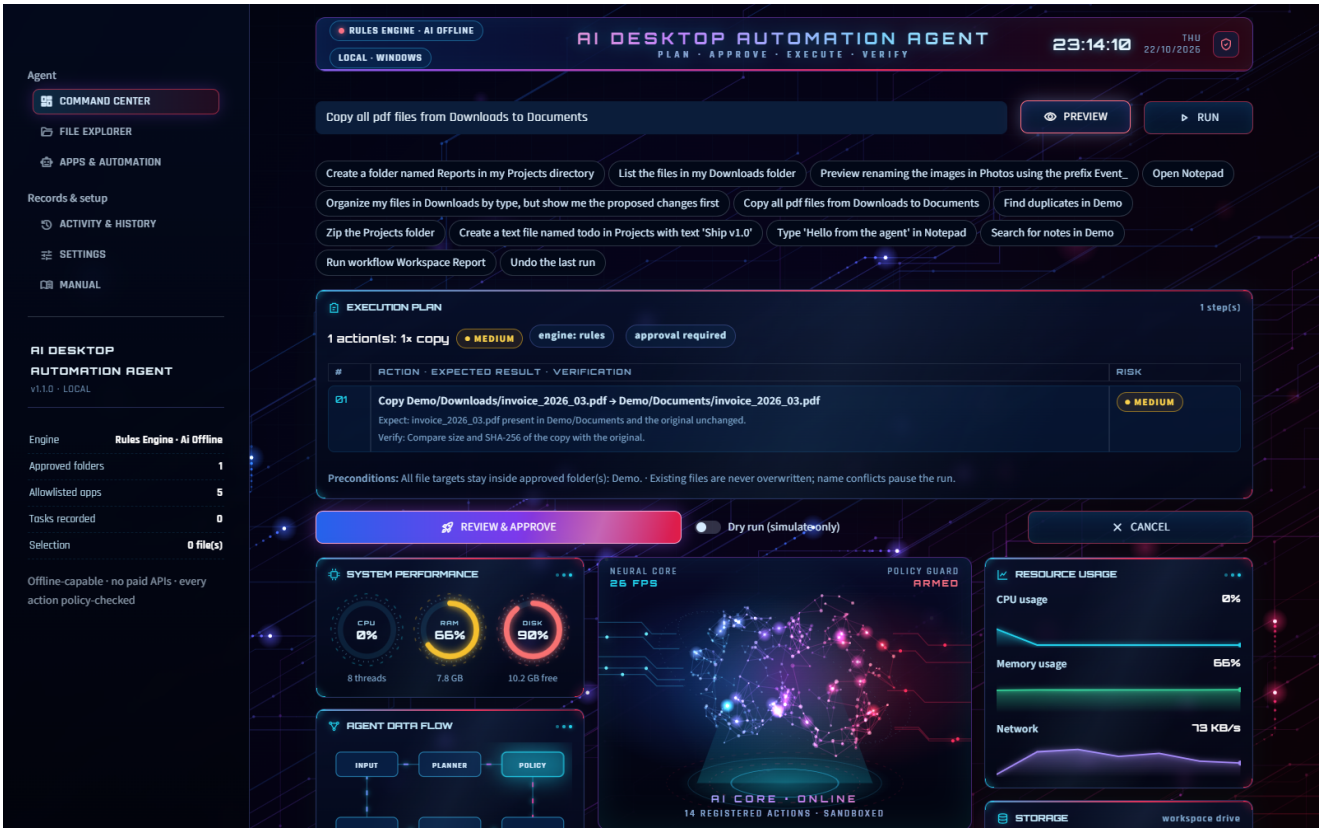
Manual alternative: python -m venv .venv, then .venv\Scripts\pip install -r requirements.txt.

4. Launching

- Double-click run.bat. The agent starts on http://localhost:8797 and your browser opens automatically. Keep the console window open while you work; close it to stop the agent.
- run.bat checks that setup.bat was run and shows a clear message if startup fails.
- Command line: .venv\Scripts\python -m app.main (same port 8797).
- Settings, history and logs are stored in %LOCALAPPDATA%\AIDesktopAutomationAgent.

5. Interface tour

Area	Purpose
Sidebar	Navigation, live status and an always-visible emergency STOP while a run is active.
Command bar	Type a task. Preview builds a plan only; Run plans and starts (approval first if needed).
Example chips	One click fills the command bar with a supported example.
Execution plan	Every step with exact targets, expected result, verification method and risk.
Execution panel	Live progress, per-step status, pause decisions and the STOP button.
System performance	Real CPU, RAM and disk gauges (psutil).
Agent data flow	Pipeline Input > Planner > Policy > Executor > Verify > Audit; the active stage glows.
AI agents	Status of the rules parser, local LLM, policy engine, executor, UI automation and audit log.
Command terminal	Live log of the current run or recent events.
Resource usage / Storage	Live CPU, memory and network charts; workspace drive usage and I/O.
Active tasks	Progress of the running plan or the most recent tasks.
Workspace intel	Files, folders, size and quarantine count across approved folders; type breakdown; largest files.
Policy shield	How many plans were approved, auto-run, dry-run, rejected, refused or undone, plus the last refusal.
Quick relaunch	Your recent commands as one-click buttons, plus 'Undo last run'.
Background & theme	Animated red/blue circuit-board background and boot sequence; switch Dual Neon / Cyber Blue or turn effects off in Settings > Workspace.



A reviewable plan with risk levels and verification steps.

6. Writing commands

Type naturally. Name folders by their approved alias (e.g. Demo) or by a subfolder name (e.g. Photos). Chain steps with 'then'. Say 'preview', 'show me first' or 'don't move anything yet' for a dry run. If something is ambiguous the agent asks a question instead of guessing.

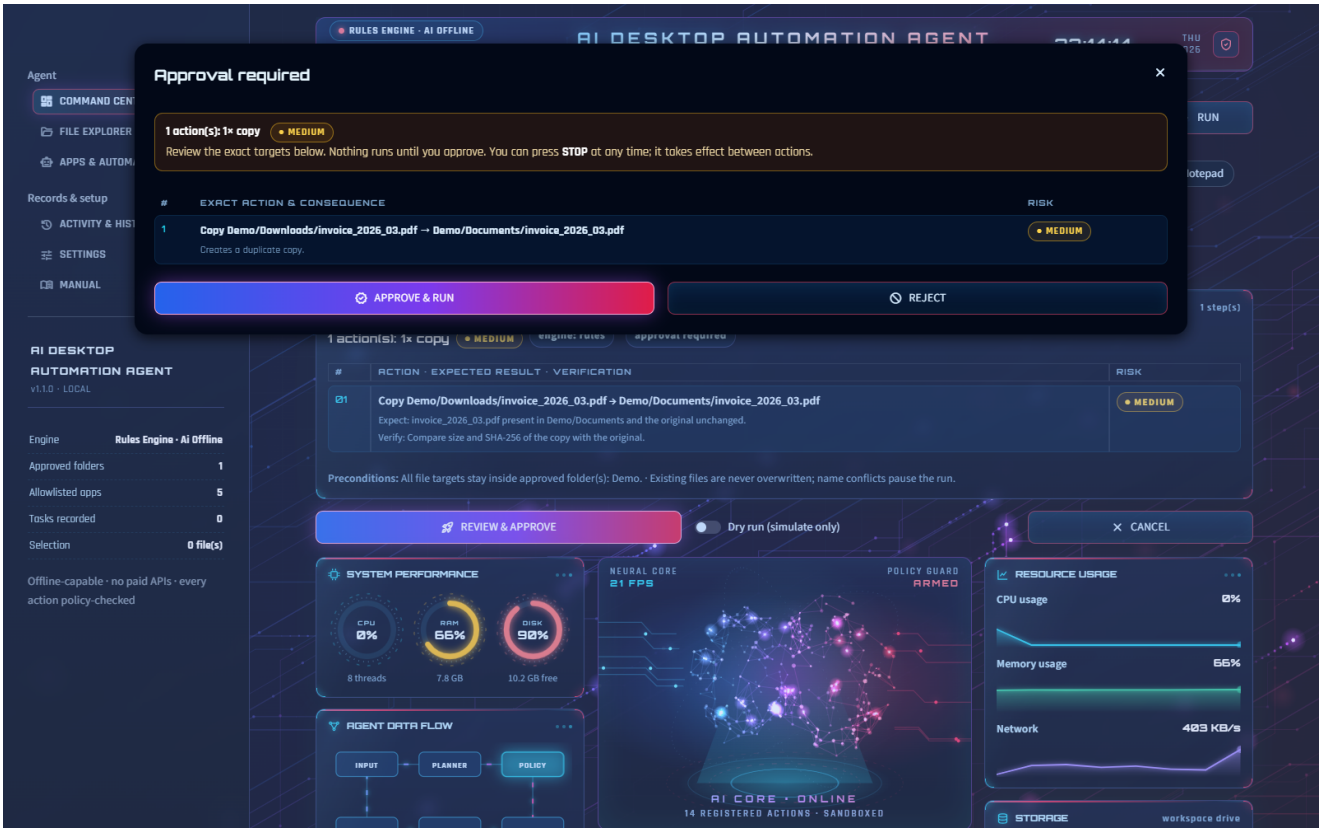
Task	Example
Create folders	Create a folder named Reports in my Projects directory
Several folders	Make folders Alpha, Beta and Gamma in Archive
List / filter	List the pdf files in Downloads | show images in Photos recursively
Search	Search for 'invoice' in Downloads
Copy / move	Copy all pdf files from Downloads to Documents | move notes.txt to Archive
Rename one	Rename roadmap.txt to plan.txt
Bulk rename	Preview renaming the images in Photos using the prefix Event_
Numbered rename	Rename the selected files to Holiday with numbers
Organize	Organize my files in Downloads by type, but show me the proposed changes first
Duplicates	Find duplicates in Demo
Zip	Zip the Projects folder
Text note	Create a text file named todo in Projects with text 'Ship v1.0'
Delete (quarantine)	Delete notes_copy.txt

Task	Example
Launch app	Open Notepad | launch calculator
Open file / folder	Open report_draft.txt | open the Photos folder
Type into app	Type 'Hello from the agent' in Notepad
Inspect UI	Inspect the Notepad window
Workflow	Run workflow Workspace Report
Selected files	Select files in File Explorer, then: rename the selected files with prefix X

7. Plans, risk and approval

Risk	Examples	What happens
LOW	List files, find duplicates, open an allowlisted app, create one folder	Runs after you press Run; still visible and stoppable.
MEDIUM	Copy, move, rename, create several folders, zip, create a file, type text	Approval dialog with exact targets (configurable).
HIGH	Delete (quarantine) files	Approval dialog + tick box + type APPROVE, and a second confirmation right before each step.
REFUSED	Send/upload/share, install software, change system settings	High-risk categories this version deliberately does not perform.
BLOCKED	Passwords/cookies, security bypass, stealth or persistence, shell commands, wiping drives	Never implemented. Refused before planning.

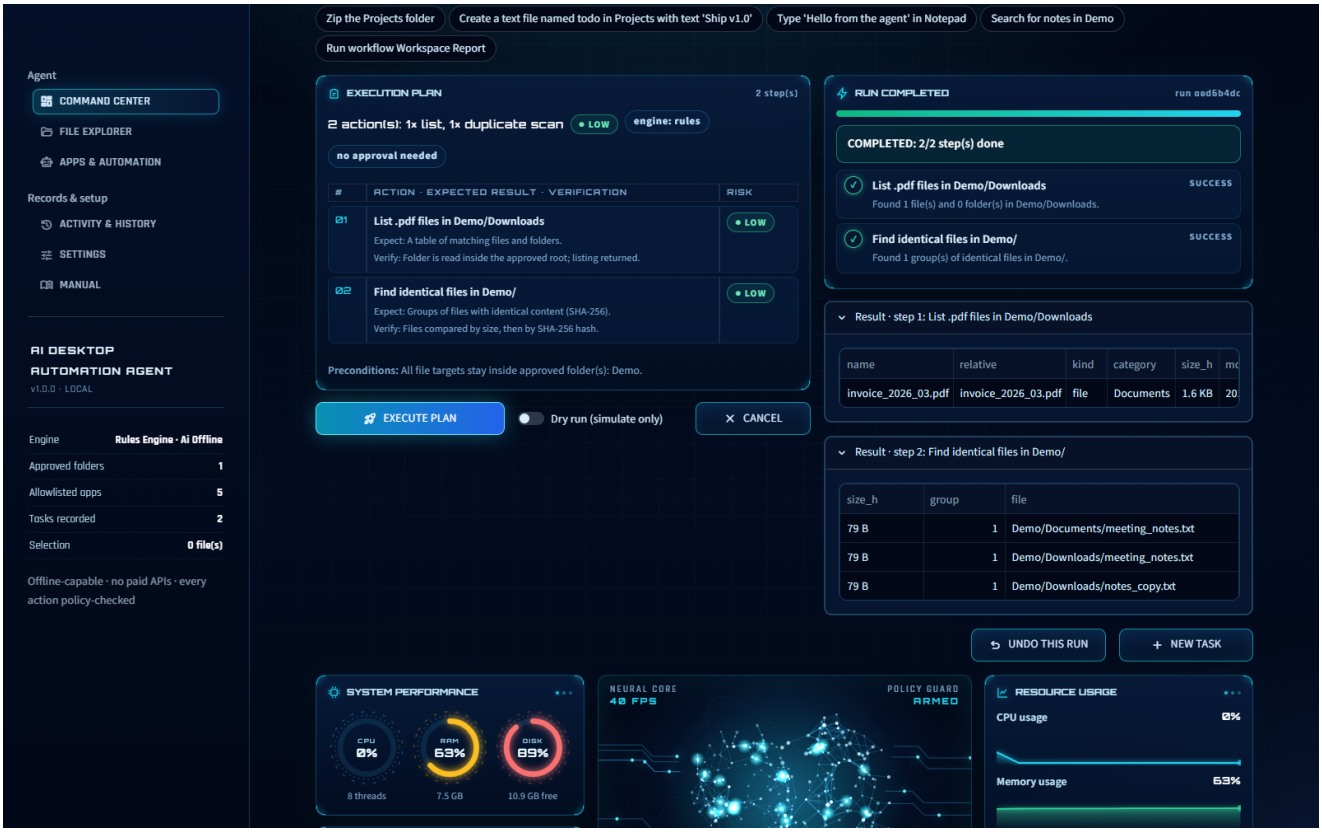
The policy engine, not the language model, decides risk and approval. A model's plan is treated as untrusted data: it is validated against a strict schema and re-checked before every action.



The approval dialog lists every exact target and its consequence.

8. Running, STOP, conflicts and dry runs

- Actions run one at a time. The STOP button (in the execution panel and the sidebar) sets a flag that is checked between actions: no new action starts after STOP. An action already in progress (e.g. copying a large file) is allowed to finish safely; the agent cannot promise to interrupt an operating-system call instantly.
- Partial completion is recorded honestly, e.g. 'CANCELLED: 2/4 steps done, 2 not run'.
- Keyboard shortcut for the sidebar EMERGENCY STOP: Ctrl+Shift+X.
- Existing files are never overwritten. On a name conflict the run pauses: skip the item, keep both (auto-rename to 'name (1).ext') or stop. A default can be set in Settings.
- Dry run simulates the plan against the planned disk state and reports what would happen. Press 'Execute this plan for real' afterwards to approve the real run.
- Each action is verified: copies by size + SHA-256, moves by source-gone/destination-present, ZIPs by integrity test, app launches by detecting the new window. Unverifiable results are marked UNVERIFIED instead of being reported as success.



Live execution with per-step verification.

9. File Explorer

- Browse approved folders, preview text files and images, and tick rows to select files.
- 'Use selection in Command Center' makes commands like 'rename the selected files ...' work.
- 'Set this folder as context' makes commands without a folder use it (stated as an assumption).
- Quick actions create, rename, copy, move, delete or organize via the same planner and approval flow.

10. Apps & Automation

- Applications: launch any allowlisted app; status shows whether it is installed.
- UI inspector: read-only list of a window's accessible controls (type, name, automation id).
- Typing: only into a window the agent opened itself in this session, located via accessibility controls, then read back to verify. If an app cannot be controlled reliably the step fails with an honest explanation.
- Workflows: save a list of commands and run them as one plan (each step still checked).

11. History, Undo and export

- Every task is stored in a local SQLite database with timestamp, request, engine, risk, approval decision, per-action results and verification flags. Error messages are redacted.

- Passwords, tokens, cookies and document contents are never stored.
- Undo reverses a task's file changes newest-first: moves and renames go back, created items go to quarantine, quarantined items are restored. It never overwrites and skips anything modified since.
- Type 'undo the last run' (or press Undo last run) to reverse the most recent file changes.
- Refused requests are logged too (status 'refused'); runs cut off by closing the app are marked 'interrupted'.
- Export as text report, CSV or JSON, download a PDF report for any single task; Clear history removes all records.

12. Settings

Tab	What you can change
Approved folders	Add/remove folders, read-only or read-write. Drive roots, the whole home folder, Windows, Program Files and AppData are refused.
Applications	Allowlist .exe programs. Shells, script hosts and system tools are refused.
Local AI	Mode (auto / rules / ollama / llamacpp), local URLs, model, timeout, test connection.
Safety & execution	Approval for medium risk, JIT confirmation, conflict default, limits, pause between actions, verification timeout.
Workspace	History on/off, animations, reset demo workspace, restore defaults.

13. Optional local AI (Ollama / llama.cpp)

- Install Ollama from ollama.com, then run once: ollama pull llama3.2:3b (internet needed for this one-time download; afterwards it runs offline).
- Settings > Local AI: keep mode 'auto' (rules first, model for phrasing the rules don't know) or choose 'ollama'. Press Test connection.
- llama.cpp: start llama-server with a GGUF model on port 8080 and choose 'llamacpp'.
- Only localhost URLs are accepted, so requests never leave your PC. Malformed or out-of-schema model output is rejected and the rules engine is used instead.

14. Safety and privacy summary

- Least privilege: only approved folders and allowlisted apps; '..', absolute paths, symlink and junction escapes are rejected after canonicalisation.
- No arbitrary execution: model or user text is never passed to eval, exec, a shell, PowerShell or Command Prompt. Only registered actions run.
- Visible operation only: no hidden windows, persistence, credential access or security changes.
- Local only: no telemetry, no cloud calls. Logs are redacted (secrets, e-mails, home path).

15. Deploying on Streamlit Community Cloud

- Push the desktop-agent folder to a GitHub repository (it already contains requirements.txt and .streamlit/config.toml).
- On share.streamlit.io choose Create app > Deploy a public app from GitHub, pick the repo, branch main, and type app.py as the Main file path.
- The hosted app runs in 'sandbox mode': every visitor gets a private copy of the demo workspace, settings and history are per session, and desktop control is disabled because a web server cannot control a visitor's PC. Use run.bat locally for the full desktop agent.

16. Troubleshooting

Problem	Fix
'Python was not found'	Install Python 3.11+ from python.org with 'Add to PATH', then rerun setup.bat.
run.bat says setup is missing	Run setup.bat first; it creates the .venv folder.
Port already in use	Another copy is running. Close it or open http://localhost:8797.
App launch 'not found'	The program isn't installed (e.g. WordPad on new Windows 11). Remove it or allowlist another.
Launch 'unverified'	The app started but no new window was detected in time; raise the timeout in Settings.
Local LLM offline	Start Ollama and pull the model, or use mode 'rules'.
Folder not found	Name it exactly as shown in File Explorer, or approve it in Settings.

17. Known limitations

- The rules engine understands the documented phrasings; very free-form requests need the optional local model.
- UI automation is limited to simple, accessible controls (e.g. Notepad's editor). Some apps expose poor accessibility data and cannot be controlled reliably.
- Opening a file with its default app cannot be verified (Windows does not report it).
- STOP takes effect between actions, not in the middle of an OS operation.
- PyInstaller packaging of Streamlit apps is possible (build_exe.bat) but larger and slower to start than run.bat; run.bat is the recommended launcher.

18. System Manager

System Manager (sidebar -> System) shows live health for the whole PC: CPU, memory, disk and battery gauges, a health score, per-core load and live charts. Tabs: Processes (search, sort, end or force-end a selected process), Storage (all drives and temp-folder size), Network (interfaces, IPs, traffic, connection states), Services & startup (read-only lists), Controls (volume, mute, media keys, 17 Windows Settings shortcuts, lock, restart, shut down, cancel shutdown) and System info.

- Every control is checked by the system policy, not by the UI or by speech. Protected Windows processes (csrss, lsass, winlogon, svchost, explorer...), other accounts' processes and the agent itself can never be ended.

- Ending a process is MEDIUM risk and needs a confirmation click; restart/shut down are HIGH risk, need an extra tick-box and run with a 60-second Windows warning that you can cancel.
- Commands are fixed argument lists with no shell; every action is written to History.
- On the hosted website the page shows the cloud container's stats in read-only mode; controls unlock when you run the agent locally on Windows (run.bat).

19. Voice Assistant (multilingual)

Voice Assistant (sidebar -> Voice) lets you speak or type commands. Pick the assistant language first: English, Urdu, Hindi, Arabic, Persian, Turkish, French, Spanish, German, Italian, Portuguese, Russian, Chinese, Japanese, Korean, Dutch, Ukrainian, Vietnamese, Gujarati or Telugu (21 options).

- Press the microphone, speak, and stop recording. Speech is recognised offline by Vosk on this machine; each language's free model (30-320 MB) is downloaded once with the Download button.
- Replies are spoken offline: Windows voices for English, and the bundled espeak-ng voices for every other language (Urdu, Hindi, Arabic and more). Turn speech off or change the rate in Voice settings.
- Urdu: speech uses the Hindustani model and is shown in Urdu script. You can also type Urdu or Roman Urdu, e.g. "awaz barhao", "battery kitni hai", "system ki halat".
- Examples: system status | CPU usage | top processes | battery | disk space | volume up | mute | open display settings | close Notepad process | lock the screen | go to history | what time is it.
- Anything else (e.g. "open Notepad", "organize my Downloads by type") is sent to the agent, which shows the plan in the Command Center for your normal review and approval.
- Hands-free mode (local PC, on by default): the agent listens to your microphone all the time, offline. Just say "open Notepad", "search YouTube for lofi", "minimize Chrome" - no clicks, no Enter. LOW-risk commands run at once; MEDIUM-risk ones are read back and run when you say "yes"; HIGH-risk ones (deletes, shutdown) always need a click on screen. Say "stop" any time. A live indicator in the sidebar shows what it hears; turn it off on the Voice page.
- Accuracy: the high-accuracy English model (124 MB) is used when downloaded; the recogniser checks up to five alternatives and picks the one that is a valid command, and fixes common slips ("note pad" -> notepad, "you tube" -> YouTube). Background chatter that isn't a command is ignored.
- Risk is always decided by the policy engine, never by the voice layer.

20. Browser, drafts, programs and windows

The agent can now work across the whole PC. Every request still becomes a plan you review, and the policy engine checks every step.

You say	What happens	Risk
open youtube.com / go to github / open github.com in Chrome	Opens the site (http/https only)	LOW

You say	What happens	Risk
search YouTube for lofi | google weather in Lahore | look up X on wikipedia	Opens the search results	LOW
draft a Gmail to a@b.com about Lunch saying See you at 1	Opens a pre-filled draft - you press Send	MEDIUM
draft a WhatsApp message to +923001234567 saying hello	Opens WhatsApp with the text filled in	MEDIUM
open vs code | open word | open chrome	Starts an installed Start-menu program	MEDIUM
list open windows | what's open	Table of window titles and programs	LOW
minimize / maximize / restore / switch to Chrome	Changes the window and reads its state back	LOW
close the Notepad window	Closes politely, like clicking X (save prompts still appear)	MEDIUM
new tab | close this tab | next tab | reload the page	Sends the browser shortcut	LOW/MEDIUM
take a screenshot (into Photos)	Saves a PNG inside an approved folder (Undo removes it)	MEDIUM

- Accuracy: typos in the first word are corrected ("opne notpad" -> open Notepad), app, program and folder names are matched to the closest name, and every correction is listed under Assumptions. File names are never guessed - you get a "did you mean ...?" question instead.
- The agent never signs in to accounts, types passwords, sends, posts, uploads or buys. Such requests are refused with a hint; drafts are opened for you to review and send yourself.
- Links that download programs, javascript:/file: links and URLs with embedded passwords are refused.
- On the website, pages and drafts appear as link windows in the web virtual desktop; program, window, tab and screenshot control work only when the agent runs locally on Windows.